



**Faculty of Engineering & Technology Electrical & Computer
Engineering Department**

ENEE4113 - COMMUNICATION LAB

Experiment #3 PRELAB

Frequency Modulation

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Extract the message signal $m(t)$

Pr Lab experiment 3:

$$s(t) = \cos(2\pi(20k)t + 6\sin(1000\pi t))$$

[1] $\beta = \frac{K_f A_m}{f_m}$, $f_m = 500 \text{ Hz}$

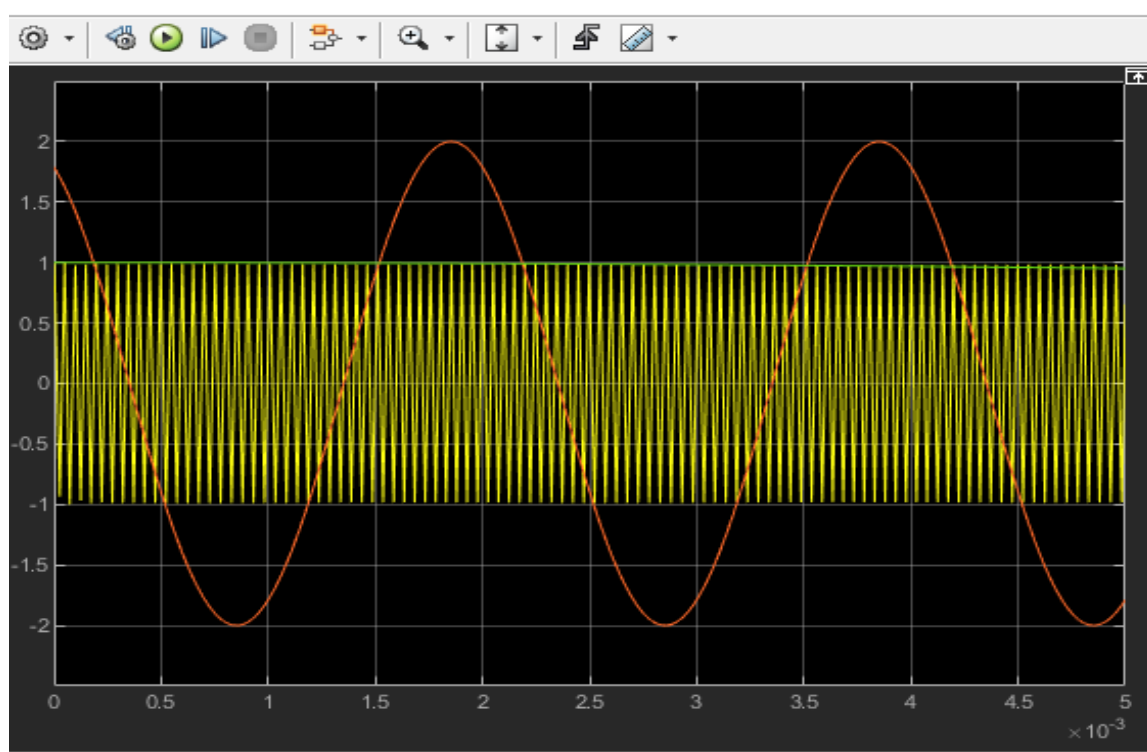
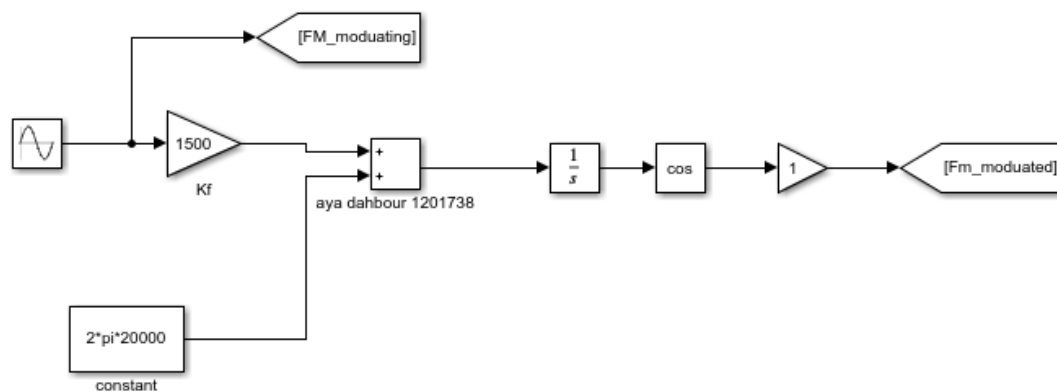
$$\Rightarrow 6 \times 500 = K_f A_m$$

$K_f = 1500$ then $A_m = 2$

$\Rightarrow$ the message signal:

$$m(t) = 2\cos(1000\pi t)$$

Plot message signal $m(t)$ and $s(t)$ versus t for $-1 \leq t \leq 1$



Extract message signal by using a phase-locked loop (PLL):

$$s(t) = \cos(2\pi(20k)t + 6 \sin(1000\pi t))$$

$$\frac{ds(t)}{dt} = - (2\pi(20k) + 6 \sin(1000\pi t)) \cdot \sin(2\pi(20k)t + 6 \sin(1000\pi t))$$

$$v(t) = 2\pi(20k) + 6 \sin(1000\pi t)$$

$$w(t) = 6 \sin(1000\pi t) \approx m(t)$$

Block diagram of the PLL process:

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    graph LR
      s_t["s(t)"] --> diff["differentiator"]
      diff --> x_t["x(t)"]
      x_t --> env["envelope detector"]
      env --> v_t["v(t)"]
      v_t --> gain["-1/-"]
      gain --> w_t["w(t) ≈ m(t)"]
  
```

